Engineering Applications of Computational Fluid Mechanics

ISSN: 1994-2060 (Print) 1997-003X (Online) Journal homepage: www.tandfonline.com/journals/tcfm20

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ENGINEERING APPLICATIONS OF COMPUTATIONAL FLUID MECHANICS
2025, VOL. 19, NO. 1, 2473992
https://doi.org/10.1080/19942060.2025.2473992

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ABSTRACT

Urban drainage models (UDMs) require observation data from sensors for calibration before managing urban floods. However, there is often the case that sensors are not available when model calibration is required, and hence sensor placing and model calibration need to be simultaneously considered in many engineering practices. While many methods are available for either sensor deployment or model calibration with given observation data, there is a lack of approach to consider these two objectives simultaneously. To address this gap, this paper proposes a method that simultaneously enables sensor placement and model calibration using Bayesian decision theory. First, a graph partition strategy is used to ensure overall uniformity in sensor distribution. Subsequently, Bayesian Experimental Design is employed to identify optimal sensor locations by maximizing expected data worth, measured by the relative entropy between prior and posterior probabilities. The UDM is sequentially calibrated using an ensemble smoother algorithm with observation data collected from these strategically placed sensors. Effectiveness and robustness of the method are tested using two real-world UDMs of different scales under various rainfall and parameter scenarios. Comparisons with two empirical approaches, where sensors are evenly deployed or installed in flood hotspots, show that the proposed method provides more accurate water level and flood predictions with less uncertainties across various scenarios. The proposed method is anticipated to be promising in engineering applications as sensor placing and model calibration are often simultaneously needed especially for many new UDMs or existing UDMs without sensors.

ARTICLE HISTORY
Received 19 November 2024
Accepted 17 February 2025

KEYWORDS
Urban drainage model; model calibration; sensor placement; Bayesian experimental design; ensemble smoother

1. Introduction

Urban areas worldwide are increasingly grappling with more frequent and intense floods caused by the insufficient capacity of existing drainage systems, a situation exacerbated by the reduction in pervious surfaces caused by rapid urbanization (de Vitry & Leitão, 2020; Wu et al., 2023). The issue is further worsened by the increases in extreme rainfall events caused by global climate change (Jamali et al., 2020). Urban floods pose substantial threats to the economy, urban water environment, and public safety (Lin et al., 2020; McCarthy et al., 2018). The ensuing societal and economic losses are now on par with those of coastal or river floods (Jiang et al., 2018). Consequently, given its potential for significant disruption to stable and sustainable urban development, the management of urban floods has become increasingly important.

To facilitate the flood prevention and mitigation, urban drainage models (UDMs) are used to simulate the complex hydraulic processes and responses of an urban system to rainfall events (Zheng et al., 2018). However, such models have many parameters, which can have considerable spatiotemporal variability (Hu et al., 2018; Zheng et al., 2023), and which must be properly calibrated to ensure adequate model performance (Vonach et al., 2019). Such calibration methods have been extensively explored, evolving from manual techniques to automated intelligent optimization methods such as gradient-based, heuristic, and combinatorial calibration (Behrouz et al., 2020; Niazi et al., 2017).

Model calibration typically involves estimating para meter values so as to align model responses with collected observation data. In practice, however, the availability of

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/19942060.2025.2473992.

observation data is often limited by the fact that sensors are generally deployed only at the terminal and/or some empirically identified locations of the drainage system, and may even be absent (e.g. in the case of new drainage systems). This situation poses significant challenges, often resulting in insufficient model reliability and inability to produce accurate and reliable predictions of the system state (Dotto et al., 2012; Her & Chaubey, 2015). Recent studies have shown that calibration using limited observation data (in response to specific rainfall events) may result in models that are unable to provide satisfactory predictions for other rainfall events or system conditions (Vonach et al., 2019; Zheng et al., 2022). Such a lack of model reliability can significantly hinder the practical application of UDMs for the prediction and management of urban floods.

In this context, some studies have attempted to quantify the implication of observation data for model calibration (Vonach et al., 2019). For instance, Freni et al. (2009) analysed the effects of data availability on UDM reliability and found that limited data availability increases the modelling uncertainty and the difficulty in identifying true parameter values. Vonach et al. (2019) tested different calibration scenarios with varying availability of observation data and verified that the number and location of observation data sources can significantly affect the effectiveness of model calibration. Therefore, increasing the number of sensors and optimizing their layout is expected to improve the effectiveness of model calibration. However, as mentioned earlier, the availability of sensors is often limited, especially in newly constructed drainage systems. Therefore, in practice, it is highly necessary to consider the installation of sensors at the same time as implementing model calibration, which can be substantially more challenging than calibration based on the use of existing sensors.

Extensive research has been conducted on sensor placement in drainage networks for purposes such as state monitoring (Wang et al., 2023), water quality management (Calle et al., 2021; Sambito & Freni, 2021) and anomaly detection (Sambito et al., 2020). However, few studies have specifically focused on sensor placement tailored for UDM calibration. The first such study was reported by Clemens (2002). Subsequently, Kleidorfer et al. (2009) proposed a stochastic modelling approach for choosing observation sites to facilitate adequate model calibration. However, their work only considered the prediction of total overflow volume using simplified hydrological models. More recently, Vonach et al. (2018) presented a heuristic approach for sensor placement in a drainage system to achieve efficient model calibration. Guo et al. (2018) used the fuzzy cluster analysis to automatically select sensor locations based on simulated time-series water level data at drainage nodes. Li (2021) combined agglomerative clustering and analysis of variance to pinpoint sensor locations for flooding hotspot detection. These studies have significantly advanced the field; however, their practical applications are limited because they assume the true or calibrated models are already known to ensure the availability of observation data across the system. Consequently, existing sensor placement approaches are still insufficient to ensure effective UDM calibration.

This paper addresses the issue of simultaneous sensor placement and UDM calibration to achieve effective and robust calibration. To this end, as suggested by Huan and Marzouk (2013), a viable option is to utilize an experimental design that strategically selects optimal sampling locations to maximize data worth for specific objectives (e.g. for model calibration, as in this paper). In general, the basis of experimental design is the characterization and quantification of expected relevant information contained in the data from a proposed design for a specific purpose. This is also sometimes referred to as 'data-worth analysis' or 'data impact' (Zhang et al., 2016). To achieve the most informative data, the information gained from different design scenarios within the design space is then maximized. Such a strategy has received increasing attention for the design of data collection in the geophysical and hydrological sciences, e.g. improving the estimation accuracy of soil hydraulic parameters (Zhang et al., 2016), and identifying groundwater contaminant sources (Zhang et al., 2015), etc. Here, it is anticipated that experimental design can be employed to optimally place sensors, providing the most informative data for calibrating model parameters and reducing model uncertainty.

To achieve this objective, and given the nonlinear nature of UDMs, the paradigm of Bayesian experimental design (Huan & Marzouk, 2013) is adopted. Relative entropy (also known as Kullback–Leibler divergence or Shannon information gain) is proposed as a quantitative measure of expected information gain from prior to posterior. Once an optimal sensor placement design is achieved, sensors are deployed at the selected locations, and real observation data are collected and used for model calibration. To maintain consistency with the Bayesian paradigm used in experimental design, this paper adopts the ensemble-based Bayesian inference for model update/calibration. Herein, the ensemble smoother with multiple data assimilation (ES-MDA) methods, widely adopted for parameter estimation in highly nonlinear problems (Emerick & Reynolds, 2013; Zhang et al., 2018) including UDMs (Huang et al., 2022), is employed to enable Bayesian inference.

The primary contributions/novelties of this work include: (i) the development of a framework tailored for simultaneously placing sensors and calibrating the urban drainage model to produce accurate and reliable model predictions; (ii) the introduction of a Bayesian experimental design method that maximizes the expected data worth of sensor networks to enhance model calibration; and (iii) the validation of the method robustness through demonstrations on two real-world UDMs of different scales, tested across various rainfall events and parameter scenarios. Note that the method's robustness refers to the effectiveness of the method under a variety of conditions, such as different UDMs, rainfall events and parameter scenarios in this paper.

This paper is organized as follows. In Section 2, the proposed methodology is described. Section 3 describes the case studies used for implementing and verifying the proposed method, followed by relevant results analysis and discussion in Section 4. Conclusions are drawn in Section 5.

2. Methodology

The proposed method makes use of an experimental design to optimize sensor placement and carry out sequential model updating for calibration. Prior to the experimental design, a graph partition strategy is implemented to ensure a uniform distribution of sensors used. This approach accounts for the fact that observation data typically only shows sensitivity towards parameters closely associated with the sensor's location. Consequently, the proposed methodology can be divided into three phases as illustrated in Figure 1.

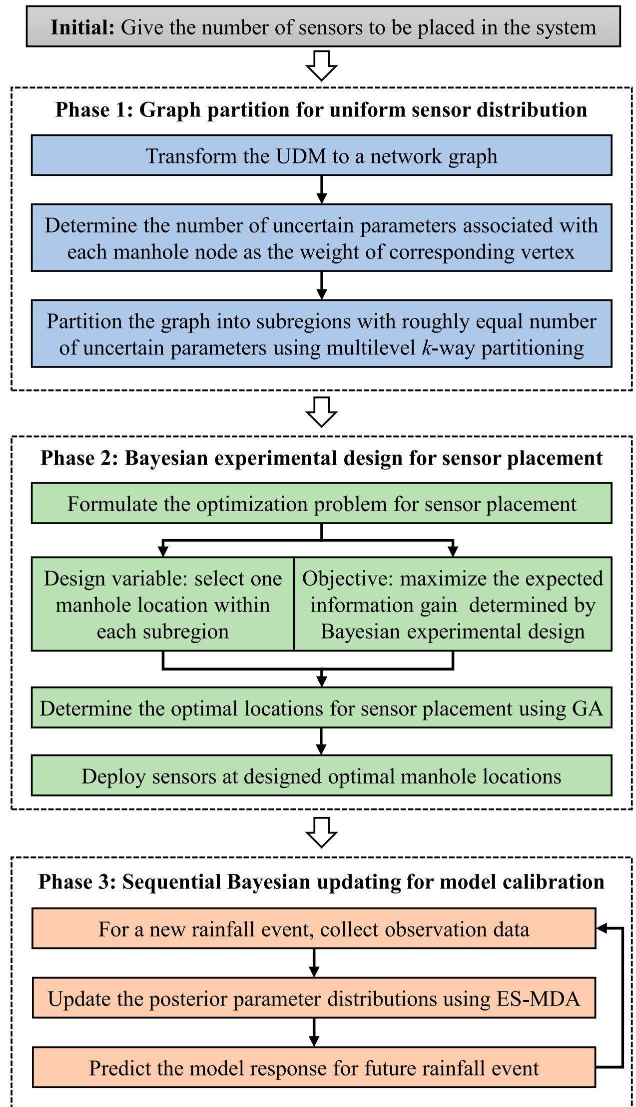

Figure 1. The proposed methodological framework for simultaneous sensor placement and drainage model calibration.

In Phase 1, the UDM is partitioned into subregions with roughly equal numbers of calibration parameters. This partitioning allows for the deployment of one sensor per subregion and is enabled by a graph-based multilevel k-way partitioning method. In Phase 2, a Bayesian experimental design approach is used to determine the expected information gain associated with a given design, and this is considered as the objective to be maximized. A Genetic Algorithm (GA) is then run to design an optimal sensor placement. Thereafter, sensors are deployed at the designed locations. Finally, in Phase 3, model calibration is conducted using the ES-MDA Bayesian model updating method to update the probability distributions of uncertain parameters from prior to posterior, after which the calibrated model is used to make predictions for future rainfall events. This process is implemented sequentially across chronological rainfall events (i.e. sequential model updating) to adapt to their temporal variation, yielding dynamically accurate models. Note that the observation data considered in this paper refers to water level data from level sensors installed at manholes, as this type of sensor is commonly used due to its stable performance, low cost, and ease of maintenance (Guo et al., 2018).

2.1. Phase 1: graph partition for uniform sensor distribution

As mentioned above, the objective of the graph partition is to divide the network into subregions of roughly equal size based on the number of calibration parameters within each subregion. It is a typical k-way partitioning problem that divides the graph into blocks of roughly equal size. Herein, the effective multilevel k-way partitioning paradigm, with extensive applications in many areas such as multi-core parallel computing, big data storage, and data mining (Buluç et al., 2016), is adopted.

To implement graph partitioning, the UDM is first transformed into a network graph, denoted as $G(V, E)$, where nodes are vertices (V) and links are edges (E). The weight of each vertex is determined by the number of calibration parameters associated with the model node. As an example, in the widely-used Storm Water Management Model (SWMM) (EPA, 2020), calibration parameters associated with a node typically include those associated with links (e.g. pipe roughness) and sub-catchments (e.g. percentage of impervious area, average slope, etc.) directly connected upstream to the node. Then the graph $G(V, E)$ with vertex weights can be partitioned using multilevel k-way partitioning with the objective that the sum of vertex weights within each subgraph is approximately equal.

The multilevel paradigm typically consists of three stages: graph coarsening, initial partitioning, and uncoar sening. This paradigm has been extensively studied in literature and widely applied in engineering; therefore, the relevant technical details are not provided here. For ease of implementation, a widely-used graph partition software package, METIS (Karypis & Kumar, 1998), was used in this work.

2.2. Phase 2: Bayesian experimental design for sensor placement

In Phase 2, on the basis of the graph partition, the sensor placement is formulated as an optimization problem. The design variable is to select one location within each subregion and the design objective is to provide the most informative observation data for model calibration; in Bayesian experimental design, this is often referred to as the expected utility (Huan & Marzouk, 2013).

Applying Bayes' theorem, the uncertain UDM parameters can be inferred from observation data collected during rainfall events. Let $\mathbf{m}$ denotes a vector of uncertain parameters, $\mathbf{r}$ a specific rainfall event, and $\mathbf{d}$ the corresponding observed outputs. The relationship between the parameters and observed outputs for a rainfall event can be expressed as:

$$\mathbf{d} = f(\mathbf{m},\ \mathbf{r}) + \boldsymbol{\varepsilon} \tag{1}$$

where $f(\cdot)$ is the composite function representing the drainage system (i.e. UDM) and $\boldsymbol{\varepsilon}$ represents the observation errors.

For a given sensor design scenario $\mathbf{s}$ (i.e. one manhole location within each subregion) and the rainfall event $\mathbf{r}$, the possible posterior distributions $p(\mathbf{m}|\mathbf{d}, \mathbf{s}, \mathbf{r})$ of model parameters conditioned on anticipated observation data $\mathbf{d}$, the design $\mathbf{s}$ and the rainfall event $\mathbf{r}$, are estimated using Bayes' theorem as:

$$p(\mathbf{m}|\mathbf{d}, \mathbf{s}, \mathbf{r}) = \frac{p(\mathbf{d}|\mathbf{m}, \mathbf{s}, \mathbf{r})p(\mathbf{m})}{p(\mathbf{d}|\mathbf{s}, \mathbf{r})} \tag{2}$$

where $p(\mathbf{m})$ is the prior distribution of uncertain parameters and reflects our knowledge about the parameters $\mathbf{m}$ before any observation data are obtained; $p(\mathbf{d}|\mathbf{m}, \mathbf{s}, \mathbf{r})$ is the likelihood; and $p(\mathbf{d}|\mathbf{s}, \mathbf{r})$ is the evidence, which can be regarded as a normalizing constant, $p(\mathbf{d}|\mathbf{s}, \mathbf{r}) \propto \int p(\mathbf{d}|\mathbf{m}, \mathbf{s}, \mathbf{r})p(\mathbf{m})d\mathbf{m}$.

The expected utility of a design $\mathbf{s}$ for a rainfall event $\mathbf{r}$ can be expressed as (Huan & Marzouk, 2013)

$$U(\mathbf{s}, \mathbf{r}) = \int_{\mathrm{D}} u(\mathbf{s}, \mathbf{d}, \mathbf{r})p(\mathbf{d}|\mathbf{s}, \mathbf{r})d\mathbf{d} \tag{3}$$

where $u(\mathbf{s}, \mathbf{d}, \mathbf{r})$ is the utility function, $U(\mathbf{s}, \mathbf{r})$ is the expected utility, and $\mathbf{D}$ is the support of $p(\mathbf{d}|\mathbf{s}, \mathbf{r})$. To enable effective evaluation of $u(\mathbf{s}, \mathbf{d}, \mathbf{r})$, relative entropy is used to measure the information gain from prior to posterior in this study, following previous work by Zhang et al. (2015). That is:

$$u(\mathbf{s}, \mathbf{d}, \mathbf{r}) = \int_{\mathrm{M}} p(\mathbf{m}|\mathbf{d}, \mathbf{s}, \mathbf{r}) \ln\left[\frac{p(\mathbf{m}|\mathbf{d}, \mathbf{s}, \mathbf{r})}{p(\mathbf{m})}\right] d\mathbf{m} \tag{4}$$

where M is the support of $p(\mathbf{m})$. Then, the expected utility $U(\mathbf{s}, \mathbf{r})$ can be rewritten by integrating Equations (2)–(4), and Monte Carlo sampling is employed for numerical evaluation (Huan & Marzouk, 2013), as follows:

$$\begin{aligned} U(\mathbf{s}, \mathbf{r}) &= \int_{\mathrm{D}}\int_{\mathrm{M}} \{\ln[p(\mathbf{d}|\mathbf{m}, \mathbf{s}, \mathbf{r})] \\ &\quad - \ln[p(\mathbf{d}|\mathbf{s}, \mathbf{r})]\} p(\mathbf{d}|\mathbf{m}, \mathbf{s}, \mathbf{r})p(\mathbf{m})d\mathbf{m}d\mathbf{d} \\ &\approx \frac{1}{n_{\mathrm{out}}} \sum_{n=1}^{n_{\mathrm{out}}} \{\ln[p(\mathbf{d}^{(n)}|\mathbf{m}^{(n)}, \mathbf{s}, \mathbf{r})] \\ &\quad - \ln[p(\mathbf{d}^{(n)}|\mathbf{s}, \mathbf{r})]\} \end{aligned} \tag{5}$$

where $\mathbf{m}^{(n)}$ are parameter samples drawn from the prior $p(\mathbf{m})$; $\mathbf{d}^{(n)}$ are data samples drawn from the likelihood $p(\mathbf{d}|\mathbf{m} = \mathbf{m}^{(n)}, \mathbf{s}, \mathbf{r})$; and n_{out} is the number of samples in this 'outer' summation. The evidence $p(\mathbf{d}^{(n)}|\mathbf{s}, \mathbf{r})$ can be also approximated using Monte Carle estimation as follows:

$$\begin{aligned} p(\mathbf{d}^{(n)}|\mathbf{s}, \mathbf{r}) &= \int_{\mathrm{M}} p(\mathbf{d}^{(n)}|\mathbf{m}, \mathbf{s}, \mathbf{r})p(\mathbf{m})d\mathbf{m} \\ &\approx \frac{1}{n_{\mathrm{in}}} \sum_{k=1}^{n_{\mathrm{in}}} p(\mathbf{d}^{(n)}|\mathbf{m}^{(n,k)}, \mathbf{s}, \mathbf{r}) \end{aligned} \tag{6}$$

where $\mathbf{m}^{(\cdot,k)}$ are parameter samples drawn from the prior $p(\mathbf{m})$ and n_{in} is the number of samples in this 'inner' summation.

The combination of Equations (5) and (6) yields an estimate of $U(\mathbf{s},\mathbf{r})$ for a specific rainfall event. Considering the large spatiotemporal variability of rainfall events, it is impossible to represent the generic rainfall characteristics via a single event. The expected utility is therefore averaged over a set of different rainfall events that can result in flood situations of concern. Accordingly, the averaged utility $U(\mathbf{s})$ for the design $\mathbf{s}$ is:

$$U(\mathbf{s}) \approx \frac{1}{n_{\mathrm{rain}}}\sum_{r=1}^{n_{\mathrm{rain}}} U(\mathbf{s},\mathbf{r}^{(r)}) \tag{7}$$

where $\mathbf{r}^{(r)}$ indicates the rainfall event r, and n_{rain} is the total number of rainfall events considered. In addition, due to the Monte Carle sampling, the experimental design requires conducting computationally intensive model evaluations for each new sample of $\mathbf{m}^{(n,k)}$, leading to a significant computational cost. To mitigate this, the same set of samples is used for both the outer summation in Equation (5) and the inner summation in Equation (6), i.e. $n_{\mathrm{in}} = n_{\mathrm{out}}$ and $\mathbf{m}^{(n,k)} = \mathbf{m}^{(n)}$; this has been found to have negligible effect on the estimation accuracy (Huan & Marzouk, 2013). As a result, only $n_{\mathrm{out}} \times n_{\mathrm{rain}}$ model evaluations are needed for the calculation of $U(\mathbf{s})$.

By maximizing the expected utility function in Equation (7), an optimal sensor location design for UDM calibration can be determined using an optimization method (such as GA). Thereafter, sensors are deployed in the field at the designed locations to collect actual observation data, thus enabling the model calibration in the following Phase 3.

2.3. Phase 3: sequential Bayesian updating for model calibration

Bayes' theorem (Equation (2)) enables us to update the posterior distributions of uncertain parameters from actual observation data in response to rainfall events. However, due to the nonlinearity of UDM, Equation (2) cannot be solved analytically. As an alternative, an approximate estimate of the posterior distribution is usually obtained for such problems. Here the ES-MDA method is adopted to approximate the posterior distributions using an ensemble of realizations (Huang et al., 2022). The ES-MDA method assimilates the same observation data multiple times by inflating the covariance matrix of observation errors, in order to improve the data match in nonlinear problems. More specifically, given the initial parameter distributions $p(\mathbf{m})$, an ensemble of N_{e} realizations is firstly generated from $p(\mathbf{m})$ for a rainfall event $\mathbf{r}$ with the corresponding observation data $\mathbf{d}_{\mathrm{obs}}$. Then the ensemble is iteratively updated from prior to posterior, conditioned on $\mathbf{d}_{\mathrm{obs}}$ with inflated observation errors, as follows:

$$\mathbf{m}_j^a = \mathbf{m}_j^f + \mathbf{C}_{\mathbf{M}D}^f(\mathbf{C}_{\mathbf{D}D}^f + \mathbf{C}_{\mathbf{D}})^{-1}[\mathbf{d}_j - f(\mathbf{m}_j^f)] \tag{8}$$

$$\mathbf{d}_j = \mathbf{d}_{\mathrm{obs}} + \sqrt{\alpha_i}\mathbf{C}_{\mathbf{D}}^{1/2}z_d \tag{9}$$

where $\mathbf{M}^f = [\mathbf{m}_1^f, \cdots, \mathbf{m}_{N_e}^f]$ and $\mathbf{M}^a = [\mathbf{m}_1^a, \cdots, \mathbf{m}_{N_e}^a]$ denote the prior and posterior ensembles of uncertain parameters, respectively; $\mathbf{C}_{\mathbf{M}D}^f$ is the $N_m \times N_d$ cross-covariance matrix between $\mathbf{M}^f$ and the predicted data, $\mathbf{D}^f = [f(\mathbf{m}_1^f,\mathbf{r}), \cdots, f(\mathbf{m}_{N_e}^f,\mathbf{r})]$; $\mathbf{C}_{\mathbf{D}D}^f$ is the $N_d \times N_d$ autocovariance matrix of $\mathbf{D}^f$; $\mathbf{C}_{\mathbf{D}}$ denotes the $N_d \times N_d$ covariance matrix of the observation errors; $\mathbf{d}_j$ is the jth realization of the observation data; and α_i is the inflation coefficient at the i^{th} iteration of update, satisfying $\sum_{i=1}^{N_{\mathrm{iter}}}(1/\alpha_i) = 1$, in which N_{iter} is the total update iterations; $z_d \sim N(0, \mathbf{I}_{N_d})$. A simple choice for α_i is $\alpha_i = N_{\mathrm{iter}}$ for all iterations (Emerick & Reynolds, 2013).

By implementing ES-MDA, an estimate of the posterior distribution $p(\mathbf{m}|\mathbf{d},\mathbf{r})$ based on collected observation data is obtained. In addition, considering the notable impact of rainfall events on model calibration as mentioned in the literature (Swathi et al., 2019), the sequential update strategy adopted here facilitates successive updating of the model parameters to obtain dynamically accurate model simulations across temporal rainfall events. That is, for a new rainfall event $\mathbf{r}^r$ and the relevant observation data $\mathbf{d}^r$, the distribution of uncertain parameters are updated from the prior $p(\mathbf{m}^{r-1})$to the posterior $p(\mathbf{m}^r|\mathbf{d}^r,\mathbf{r}^r,\mathbf{m}^{r-1})$. The evolution of $p(\mathbf{m}^r|\mathbf{d}^r,\mathbf{r}^r,\mathbf{m}^{r-1})$ based on new observation data enables a sequential model update and changing state of knowledge of the UDM. Meanwhile, the update $p(\mathbf{m}^r|\mathbf{d}^r,\mathbf{r}^r,\mathbf{m}^{r-1})$ is used to predict the model response for future rainfall events.

2.4. Empirical methods for performance comparison

To investigate and analyse the utility of the proposed method for sensor placement and calibration of the UDM, comparisons are made with two empirical approaches. The first approach, referred to as **SED** (Sensors Evenly Distributed), involves evenly distributing sensors across the drainage network. This approach is straightforward and ensures broad coverage of the area, helping to minimize information redundancy when sensor locations are close together. Due to its simplicity and computational efficiency, the **SED** method is commonly used in engineering practice, especially when no specific information is available to guide sensor placement. However, its primary limitation is that it does not prioritize sensor placement based on areas most relevant to flood or hydrological events, which may reduce its effectiveness in capturing critical data for model calibration.

The second approach, referred to as **SFL** (Sensors at Flooding Locations), strategically deploys sensors at locations with a relatively high likelihood of flooding (i.e. locations prone to flood). This approach targets critical flood-prone areas, where data is particularly valuable for model calibration. While the **SFL** approach focuses on high-priority zones, it can lead to information redundancy when sensors are placed too close together and may overlook important data from regions outside these zones. Additionally, it relies on prior knowledge of flood likelihood, which might not always be accurate or available.

These two approaches have been typically implemented in engineering practice (Vonach et al., 2019), given the lack of a state-of-the-art method for simultaneously considering sensor placement and model calibration. While many evolutionary algorithms (such as GA) have been applied to this problem, they have focused solely on either the sensor placement with already calibrated model parameters (Banik et al., 2017), or on the model calibration problem with fixed sensor placement (Niazi et al., 2017).

The SED approach can be achieved by implementing the graph partition in Phase 1 of the proposed method and simply placing sensors at the terminal of each partition. The SFL approach involves evaluating flood likelihood using Monte Carlo model simulations for rainfall events considered for experimental design, based on prior parameter distributions. The locations with maximum expected flood volumes are then considered as sensor locations. In the discussion that follows, our proposed method based on Bayesian experimental design is referred to as **BED**. To enable a more complete performance comparison, the sequential Bayesian updating is also conducted using sensors designed by the SED and SFL methods.

3. Case studies

3.1. Description

The feasibility and application of the proposed method are demonstrated using two real-world UDMs derived from an open-access Bellinge data set shared by Pedersen Nedergaard et al. (2021), in which a large-scale SWMM of the real urban drainage system in Bellinge, Denmark, as well as recorded rainfall data, are provided. For analysis convenience, two parts of this real-world model (i.e. the area enclosed by the dashed line in Figure 2) were extracted to create two individual UDMs. These were formed by disconnecting the selected parts from the entire model and adding free outfall nodes at the terminals, as depicted in Cases 1 and 2 in Figure 2. Case 1 comprises 38 sub-catchments, 54 nodes and 53 conduits, while Case 2 consists of 113 sub-catchments, 132 nodes and 132 conduits. The two case studies differ in their complexity in terms of network topology, scale, and hydrological conditions. Specifically, Case 1 represents a smaller network with a simpler topology, whereas Case 2 involves a larger, more complex network with an intricate topology and more variable hydrological conditions, including higher discharge and flow complexity. These differences are leveraged to validate the effectiveness of the proposed method.

For both cases, a total of 12 types of model parameters were calibrated (see Table 1). This includes almost all the parameter types considered to be calibrated in relevant approaches (Niazi et al., 2017). Among these, 11 parameter types are associated with the conceptual sub-catchments while the last type relates to the conduits of the drainage network. All types are either difficult, or impossible, to measure directly, or are spatiotemporally variable and must therefore be calibrated. Other model parameters, such as conduit sizes, are assumed to be known constants. The ranges of calibration parameters (see Table 1) are based on the literature (Rossman, 2015; Swathi et al., 2019). Note that m_0 refers to the initial guessed values for the 'width' parameter of each sub-catchment. Considering its empirical nature and variation for different sub-catchments, these values are typically estimated by experienced modellers, and the 'width' range is given as $\pm 50\%$ of m_0 instead of a fixed range. Consequently, there are 471 and 1375 unknown parameters for Cases 1 and 2, respectively, and their prior distributions are assumed to be uniform within the ranges given in Table 1. It is worth noting that while reducing parameters through methods like sensitivity analysis is a common strategy to improve calibration efficiency, this study considers all uncertainty parameters and leverages the advantages of the ES-MDA method for uncertainty analysis, providing a more robust and comprehensive calibration framework. Similar approaches have been validated in recent studies, such as Huang et al. (2022) and Yao et al. (2025).

The unknown parameters are calibrated using the proposed method including Bayesian experimental design and sequential Bayesian updating following the procedures in Figure 1. To approximate real conditions, the rainfall data set provided by Pedersen Nedergaard et al. (2021) was used and 12 recorded rainfall events from 2011 to 2012 were selected to enable the implementation of the proposed method, as listed in Table 2. These rainfall events were selected because they can lead to diverse flood situations. Specifically, rainfall events H1–H5 that occurred in 2011 are considered as historical events to calculate the average expected utility in

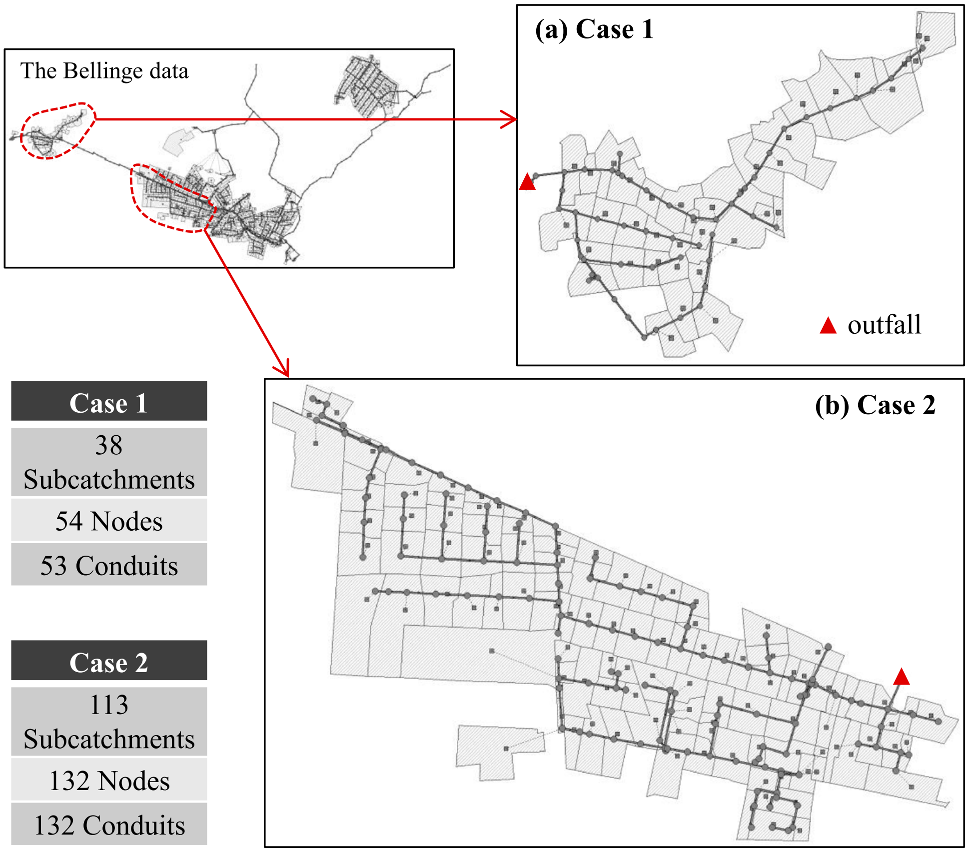

Figure 2. Layouts of urban drainage system models for the two studied cases.

Table 1. Uncertainty parameters for model calibration.

Parameters	Units	Descriptions	Intervals
Width	m	Width of overland flow path	[$0.5m_0$, $1.5m_0$]
%Slope	%	Average surface slope	[0.01, 5]
%Imperv	%	Percent of impervious area	[0, 100]
N-Imperv	–	Manning coefficient for Impervious area	[0.01, 0.04]
N-Perv	–	Manning coefficient for pervious area	[0.1, 0.8]
Dstore-Imperv	mm	Depth of depression storage on impervious area	[0.2, 5]
Dstore-Perv	mm	Depth of depression storage on pervious area	[2, 10]
%Zero-Imperv	%	Per cent of impervious area with no depression storage	[0, 100]
MaxRate	mm/h	Maximum rate on Horton infiltration curve	[20, 80]
MinRate	mm/h	Minimum rate on Horton infiltration curve	[0, 10]
DecayCoeff	1/h	Decay constant for the Horton infiltration curve	[2, 7]
Roughness	–	Manning's roughness coefficient for conduit links	[0.01,0.03]

Bayesian experimental design (i.e. Equation (7)); then the designed sensors are deployed to collect observation data to enable the sequential Bayesian updating across rainfall events 1–7. As shown in Table 2, the 12 selected rainfall events have mean intensities ranging from 1.5 to 24.9 mm/h and varying peak intensities, total depths, and durations, hence representing a wide range of rainfall characteristics. Therefore, they are well-suited for the design of effective sensor placement and model calibration as performed in this paper.

As, given the nature of these study cases, there are no actual sensors being installed, simulated observation data are generated by introducing observation errors (i.e. noise) to the corresponding model simulation outcomes based on assumed true values for the model parameters. Accordingly, observation errors $\boldsymbol{\varepsilon} \sim N(0, 0.1)$, with a unit of m, were assumed for the drainage system water level sensors. Further, to evaluate broad effectiveness of the proposed method, five different 'true' parameter scenarios (P1–P5) were randomly generated. Table 2 reports the flood volumes corresponding to various rainfall events under these five parameter scenarios. More detailed information is provided under 'Data Sets S1–S10' in the Supplementary material.

Table 2. Key characteristics of the selected rainfall events.

Events	Start time	End time	Total rainfall (mm)	Intensity (mm/h)		Flood volume in Case 1 (m^3)					Flood volume in Case 2 (m^3)				
				Mean	Max	P1	P2	P3	P4	P5	P1	P2	P3	P4	P5
H1	2011/6/6 5:15	2011/6/6 9:47	7	1.5	72	103	108	19	53	71	147	82	83	109	137
H2	2011/7/9 14:13	2011/7/9 17:56	12	3.1	84	446	476	115	379	395	815	528	569	627	821
H3	2011/8/6 18:16	2011/8/6 18:31	5	20.8	60	59	66	12	25	41	81	40	44	60	70
H4	2011/8/26 21:31	2011/8/27 1:04	13	3.7	120	506	550	147	447	457	954	650	709	757	948
H5	2011/9/4 22:11	2011/9/5 2:11	15	3.7	72	346	365	64	264	283	531	342	277	408	596
1	2012/6/29 4:33	2012/6/29 8:03	19	5.5	108	578	650	176	536	557	1248	797	766	960	1316
2	2012/7/14 16:04	2012/7/14 18:18	9	3.9	96	245	263	55	187	202	388	266	249	283	392
3	2012/7/16 7:51	2012/7/16 11:25	9	2.5	36	81	95	13	47	61	168	74	63	110	180
4	2012/7/27 22:37	2012/7/27 23:18	11	16.7	84	434	468	108	367	399	830	535	566	619	833
5	2012/7/30 16:33	2012/7/30 16:46	5	24.9	84	104	105	23	46	76	137	82	91	101	121
6	2012/8/6 4:39	2012/8/6 9:02	11	2.5	36	145	158	23	109	113	253	140	124	181	276
7	2012/8/22 3:02	2012/8/22 3:51	7	9.1	96	96	104	20	39	70	176	79	80	106	171

3.2. Method application

The graph partition in Phase 1 was implemented using the multilevel k-way partitioning method embedded in METIS Version 5.1.0 (Karypis, 2013). Note that the options of forcing contiguous partitions and identifying the connected components were selected to improve the effectiveness and efficiency of the graph partition.

For the Bayesian experimental design in Phase 2, choosing an appropriate sample size is crucial for balancing convergence and computational efficiency. To illustrate this, Figure 3 is presented, comparing two random design schemes with the optimal design obtained by the BED method. The graph shows how the expected utility changes with sample size, alongside the corresponding computational time. Additionally, parallel tests were performed for each design scheme, and the results showed minimal fluctuation in expected utility values, indicating stable outcomes. It can be generally observed that as the sample size increases, the expected utility improves, but computational time grows exponentially. Therefore, selecting a large sample size to achieve convergence of the expected utility would significantly increase the computational burden, making it inefficient in terms of computational cost. Importantly, the relative expected utility values between different designs show clear differences even at smaller sample sizes (e.g. in the thousands), ensuring that the optimal design can be reliably identified without the need for excessively large samples. Considering these factors, a sample size of 2000 was selected for both Cases 1 and 2, providing a practical trade-off between computational efficiency and reliable optimization.

For the optimization task of the experimental design, a self-implemented version of GA was developed to find sufficiently good near-optimal design solutions. Details on the principle of GA can be found in many references (e.g. Niazi et al., 2017), which therefore are not repeated here.

To simplify the implementation, the number of ensemble realizations for ES-MDA in Phase 3 was uniformly set to $N_e = 2000$ for both cases. Other settings of the ES-MDA method were determined through trial-error tests following practices in the literature (e.g. Emerick & Reynolds, 2013; Zhang et al., 2018). Specifically, the number of iterations was set to $N_{\text{iter}} = 5$, and the inflation coefficient was defined as $\alpha_i = N_{\text{iter}}$.

3.3. Performance evaluation

To evaluate the performance of the proposed method, the corresponding assessment metrics for output state variables (i.e. water level at the manhole for the two

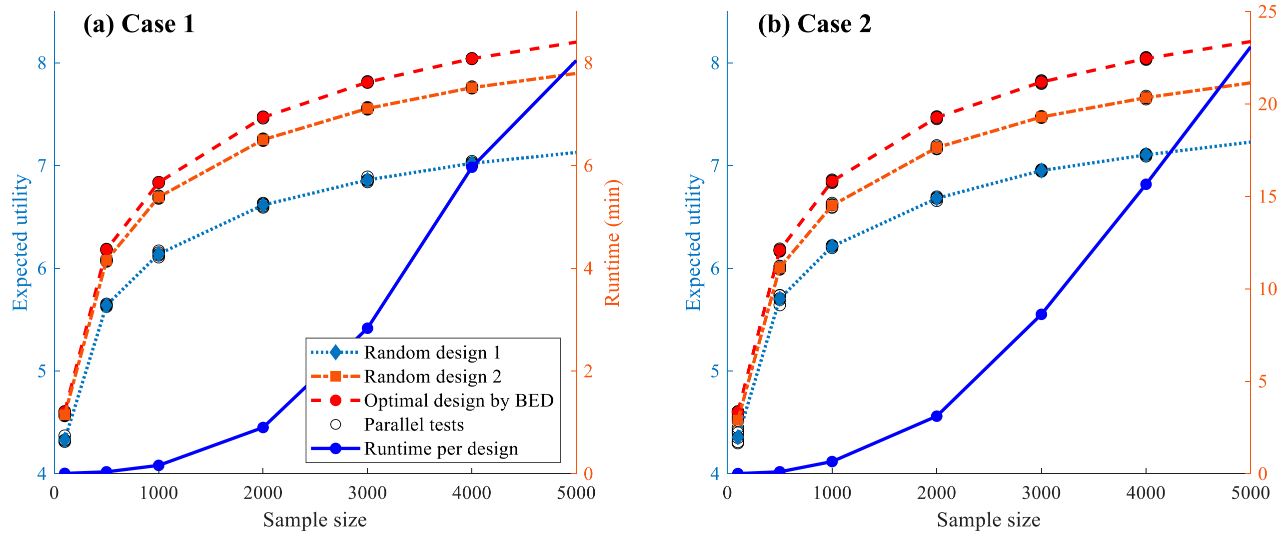

Figure 3. Convergence of expected utility and computational time for different design schemes across sample sizes in (a) Case 1 and (b) Case 2.

cases) and uncertainty parameters were used. Given that ensemble-based Bayesian updating generates a collection of estimates that describe probability distributions, the estimation accuracy and associated uncertainty were quantified individually, as follows:

- For the estimation of state variables, the Root Mean Square Error (RMSE) of the mean estimates and the mean value of standard deviations (SD) of the estimates across the time profile for each state variable, referred to as **MRMSE** and **SDM** respectively, were calculated to represent the estimation accuracy and uncertainty.
- For parameter estimation, the relative bias (RB) of each estimate for each parameter was calculated and then the statistical mean and standard deviation of all RB values for each parameter, referred to as **RBM** and **RBSD**, were used to represent the estimation accuracy and uncertainty, respectively. Here RB was calculated as $\mathrm{RB} = |m - m_{\mathrm{true}}|/(m_2 - m_1)$, in which the absolute deviation $|m - m_{\mathrm{true}}|$ is divided by the parameter interval $(m_2 - m_1)$ for normalization (Zheng et al., 2018).

Therefore, smaller values of MRMSE and SDM indicate more accurate state estimation with less uncertainty, and smaller values of RBM and RBSD indicate more accurate parameter estimation with less uncertainty. In addition, observation data with water levels less than 0.1 m were removed from the calculation of assessment metrics as these tiny values of water level are generally not of interest in practice.

4. Results and discussions

4.1. Case 1: placing five sensors

For Case 1, the placement of five sensors for model calibration was first applied as an example to demonstrate the implementation procedure and performance of the proposed method.

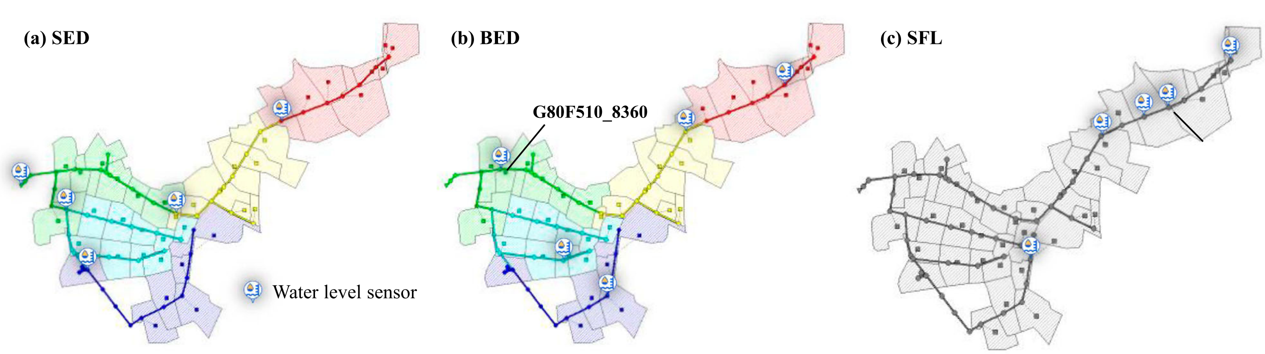

Figure 4. Sensor locations designed by the SED, BED and SFL methods, respectively.

4.1.1. Optimal sensor placement

The sensor locations obtained by the SED, BED and SFL methods are given in Figure 4. The different colours for sub-catchments, conduits and nodes in Figure 4(a,b) indicate the partitioned subregions in Phase 1. As shown in Figure 4(a), the SED method provides 5 evenly distributed sensors located at terminal nodes of subregions, collecting the superposed signals. In contrast, the BED method produces optimized sensor locations within each subregion, which differ from those produced by the SED method, as shown in Figure 4(b). Meanwhile, the SFL method concentrates the sensors in areas with severe flood likelihood, such as the 4 sensors in the upper right of the UDM, as indicated in Figure 4(c). As can be seen, compared to the other two empirical methods (SED and SFL), the BED method can achieve an optimized sensor placement that considers both distribution uniformity and data worth maximization. While the distribution appears even, the key advantage of BED is its ability to strategically place sensors in areas where they provide the most valuable data for model calibration, taking into account flow dynamics and system behaviour. This targeted approach is expected to lead to a more effective sensor network for model calibrations, beyond simply distributing sensors evenly, as in the SED method, or focusing solely on flood-prone areas, as in the SFL method.

4.1.2. Model calibration and prediction performance

Before assessing performance, the effectiveness of the proposed approach to model calibration via sequential Bayesian updating is examined. As an example, the results obtained using sensor networks designed by the BED method are utilized, with parameter scenario P1 serving as a reference.

Figure 5 presents boxplots of assessment metrics for sequential Bayesian updating of water level state across rainfall events 1–7 (see Table 2) at sensor locations and for all nodes. The first (leftmost) boxes in each of Figure 5(a–d) correspond to initial water depth estimates based on the initial ensemble of parameter realizations (see the first/leftmost boxes in Figure 6(c,d)). Figure 5(a,b) shows that the updated state estimates at the sensor locations are generally more accurate, and with less uncertainty, than the results initially predicted for each rainfall event – the boxes marked as 'updated' are lower than those marked as 'predicted' for both of the metrics MRMSE and SDM. This pattern is also observed for the state estimates of all nodes in Figure 5(c,d). These results indicate that the accuracy and uncertainty of state estimates are both improved after Bayesian updating, demonstrating the effectiveness of the proposed method of sequential updating using ES-MDA. In addition, the distinct differences between updated and predicted state estimates along the sequential updating trajectory suggest that the UDM calibration process is somehow sensitive to rainfall events, as has also been reported in the literature (Swathi et al., 2019). The results support the notion that sequential model updating (as proposed in this study) is necessary to maintain the dynamical effectiveness of UDM models.

Figure 6 further illustrates the parameter calibration results during the sequential updating process using ES-MDA. Specifically, Figure 6(a,b) provides an example of the iterative and sequential updates of the probability distribution for the %Imperv parameter of sub-catchment G80F510_8360, respectively, with the true parameter value indicated (black dotted line) and the sub-catchment's location marked in Figure 4(b). Figure 6(a) shows the evolution of the parameter distribution during the iterative calibration process for rainfall event 1. Starting from a prior uniform distribution, the distribution becomes progressively more concentrated as observational data are assimilated, resulting in a posterior distribution with reduced uncertainty. After five iterations, the final posterior distribution exhibits a sharper peak, and the parameter value corresponding to the maximum probability is closer to the true value. This illustrates the effectiveness of the ES-MDA method in iteratively refining parameter estimates. Figure 6(b) extends this analysis by showing the sequential updates of the same parameter across rainfall events 1–7. As additional observational data from subsequent rainfall events are incorporated into the calibration process, the posterior distribution continues to narrow, reflecting reduced uncertainty in the parameter estimates. This progressive refinement highlights the strength of the sequential Bayesian updating approach in improving parameter estimates over time.

To evaluate the overall calibration performance, Figure 6(c,d) presents boxplots of the assessment metrics RBM and RBSD for all parameters across rainfall events 1–7. The results reveal a consistent reduction in uncertainty (as shown by decreasing RBSD values) as more rainfall events are included in the calibration process. This is attributed to the availability of additional calibration data, which constrains the parameter distributions more effectively. However, the accuracy of parameter estimation (evaluated with RBM) does not show a significant improvement with model updating through the seven rainfall events. This suggests that the identification of 'true/best' values for parameters is challenging. Note, however, that instead of identifying the optimal parameter set, the updating procedure tends towards a range of feasible parameter sets that match the data collected at sensor locations equally well. This is the

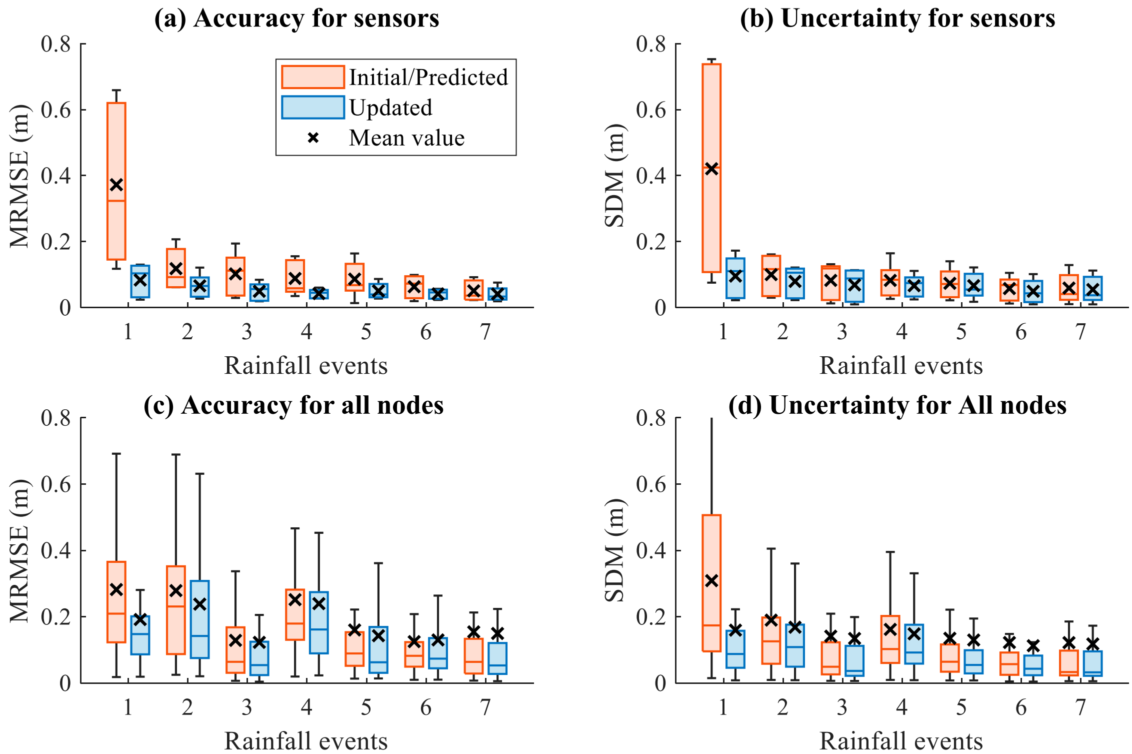

Figure 5. Assessment metrics for predicted and updated water levels at (a,b) sensor locations and (c,d) all nodes across rainfall events 1–7.

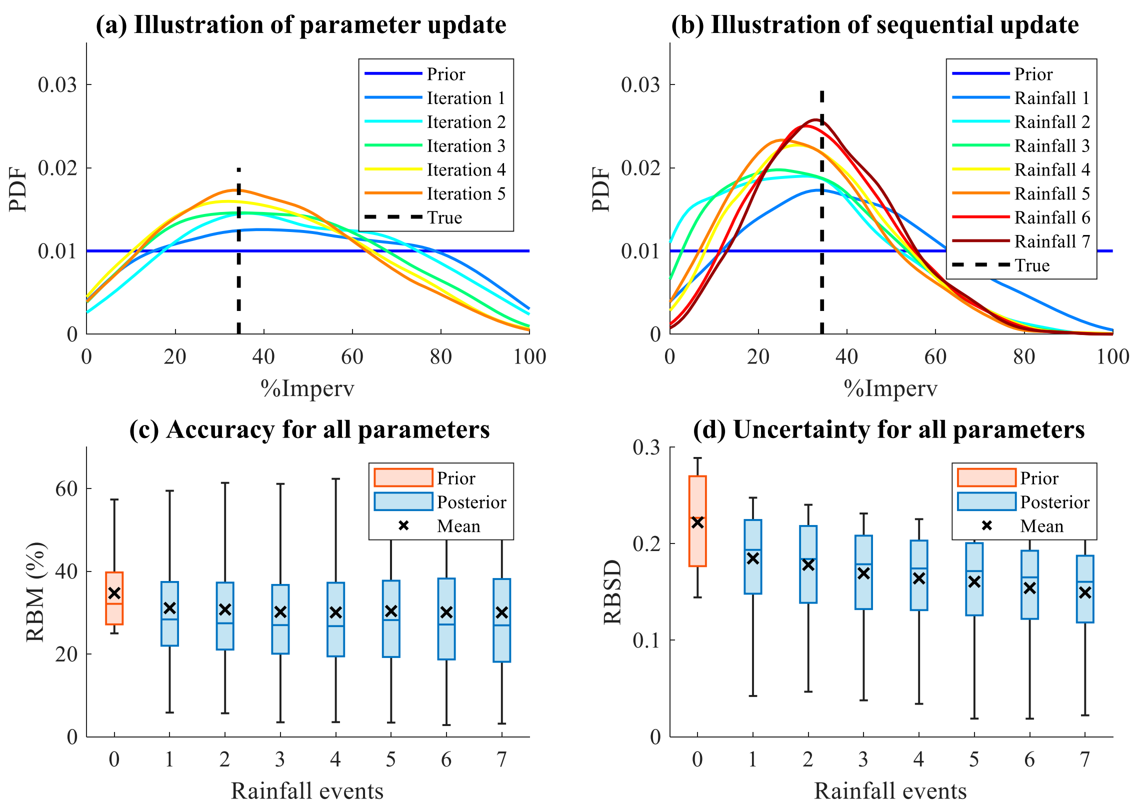

Figure 6. Results of parameter calibration, including: (a) iterative updates for the parameter %Imperv of sub-catchment G80F510_8360 during rainfall event 1, (b) sequential updates across rainfall events 1–7 for the same parameter (location marked in Figure 4(b)), and (c,d) boxplots of assessment metrics (RBM and RBSD) for all parameters.

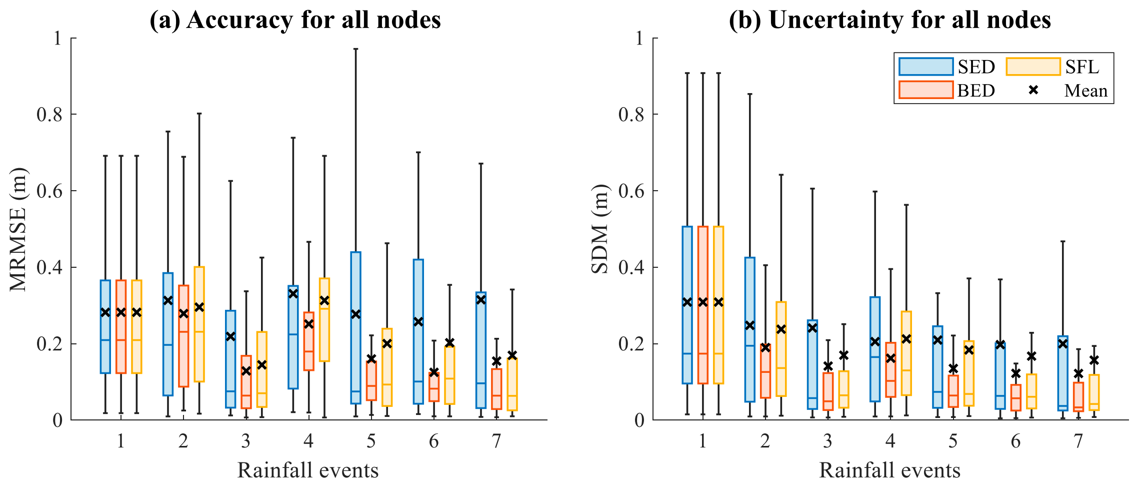

Figure 7. Comparison of assessment metric values for water level predictions at all nodes for three design methods.

equifinality issue widely encountered in many hydrological and hydrodynamical problems. The approach presented in this study acknowledges the inherent uncertainty and non-uniqueness of parameter estimation for complex UDMs. In practice, effective output/state estimation (e.g. flood prediction) is often of greater concern than precise parameter identification, as robust predictions of system behaviour are more critical for decision-making. The aim is to identify parameter sets that lead to reliable predictions while accounting for the variability and uncertainty in the system. Further discussion on parameter-identifiability for UDM calibration can be found in Huang et al. (2022).

Having verified the effectiveness of sequential model updating, the prediction performance of calibrated models using the SED, BED and SFL methods is then compared and demonstrated. Figure 7 compares metric values for all nodes across rainfall events 1–7 under parameter scenario P1, showcasing the prediction performance for the entire model domain. As shown by the box plots and mean values, the BED method consistently achieves lower metric values for both MRMSE and SDM across rainfall events 2–7, followed by SFL and then SED. This suggests that the BED method results in calibrated models with greater prediction accuracy and less uncertainty compared to the SFL and SED methods, making it the most effective for providing informative observation data for UDM calibration. Additionally, note that the initial water level predictions for rainfall event 1 are identical across the three methods (see the leftmost boxes in Figure 7(a,b)), as the model has not yet been calibrated.

4.1.3. Flood prediction performance

While overall model performance is important, practical applications often emphasize the prediction performance of UDMs under flood conditions. In urban flood management, state changes in water level and flood discharge at flood/overflow manhole nodes, which signal the timing and intensity of floods, are particularly crucial. Therefore, a detailed analysis is conducted to focus on the prediction performance of calibrated models in flood scenarios.

As an example, the flood prediction performance for rainfall event 4 under parameter scenario P1 is firstly evaluated, which represents the most severe flood situation (i.e. the largest flood volume) among events 2–7 (see Table 2). Figure 8 displays the absolute errors (AEs) between predicted and actual flood volumes for various nodes across the system, based on calibrated models from the three design methods. Larger circles denote greater prediction errors. The figure also indicates total and maximum prediction errors for comparison. As can be seen, the spatial distribution of flood prediction errors differs noticeably among the three different models, with the BED method showing relatively smaller errors (smaller circles in Figure 8(b) compared to Figure 8(a,c)). This result is also evident from the total and maximal error values indicated in the figure. For instance, the total AE for the BED method is 108 m^3, much lower than 156 m^3 for the SED method and 166 m^3 for the SFL method, against an actual flood volume of 434 m^3. These findings suggest that the calibrated model using the BED method provides better prediction accuracy for rainfall event 4, both in spatial distribution and overall performance.

To better illustrate flood prediction performance, the time-series profiles of predicted water levels and flood discharge were analysed at a representative flood node, G80F461, as shown in Figure 9. The location of this node is shown in Figure 8(b). Note that this node is not a monitoring point in any of the three design schemes, so its prediction results can partially represent the predictive performance of calibrated models at non-observed

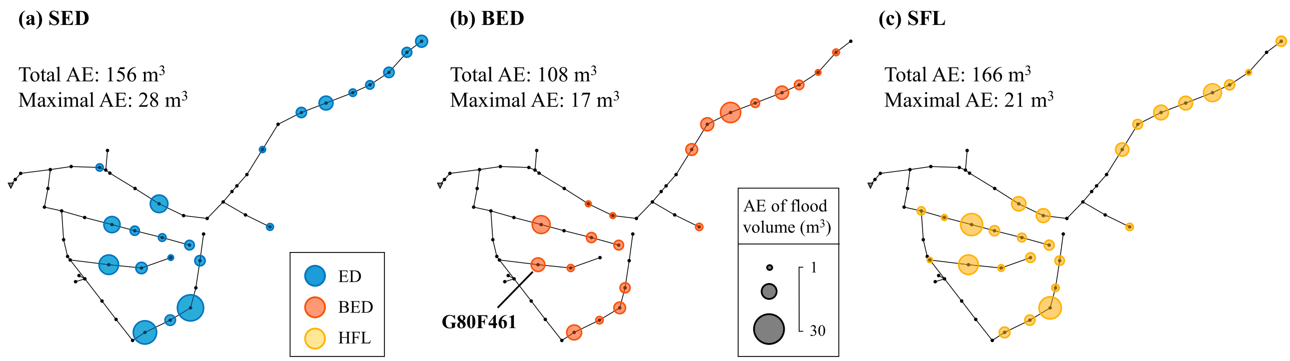

Figure 8. Layouts of absolute errors (AE) for predicted versus actual nodal flood volumes corresponding to rainfall event 4 for three design methods.

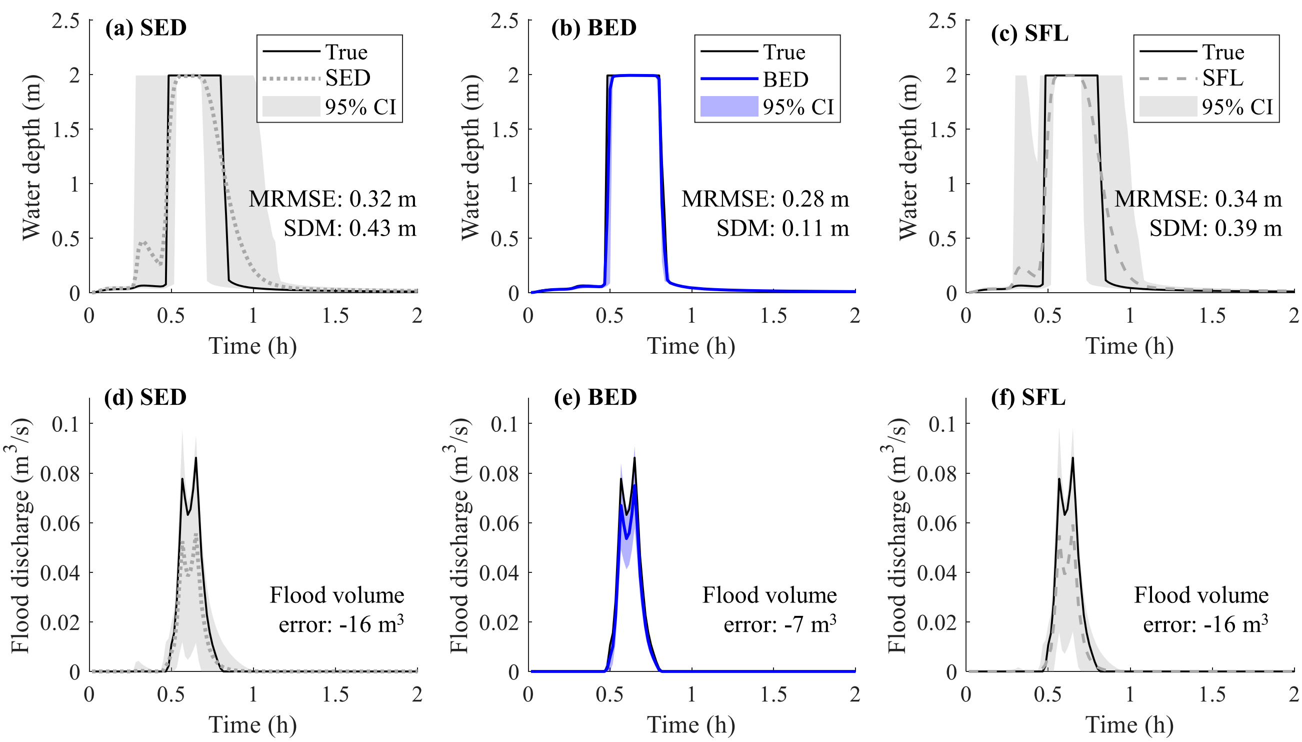

Figure 9. Comparison of water depth predictions (a–c) and flood discharge predictions (d–f) at node G80F461 corresponding to rainfall event 4 for three design methods. The shaded regions represent the 95% confidence interval (CI), derived from the ensemble of the model predictions. The node location is marked in Figure 8(b).

locations. Figure 9 also presents the MRMSE and SDM metrics for predicted water levels and the errors between predicted and actual flood volumes. The shaded regions in Figure 9 represent the 95% confidence interval (CI), calculated from the ensemble of the model predictions. As shown in Figure 9, both the water level and flood discharge profiles predicted by the BED method exhibit fairly accurate data-matching, while those predicted by the SED and SFL methods deviate significantly from the true profiles. This is also evident in the MRMSE and SDM values and flood volume errors shown in Figure 9. These results suggest that the BED method outperforms the SED and SFL methods in both water level and flood discharge predictions.

To further verify the above finding, Figure 10 presents a performance comparison for flood prediction across the entire model domain. Specifically, the prediction accuracy and uncertainty for water levels and flood volumes at all flood nodes were given in Figure 10(a,b) and Figure 10(c,d), respectively. As shown in Figure 10, for rainfall events 2–7, the BED method consistently produces the smallest box plots and mean values for water level and flood volume predictions at flood nodes, followed by the SFL method, which generally outperforms

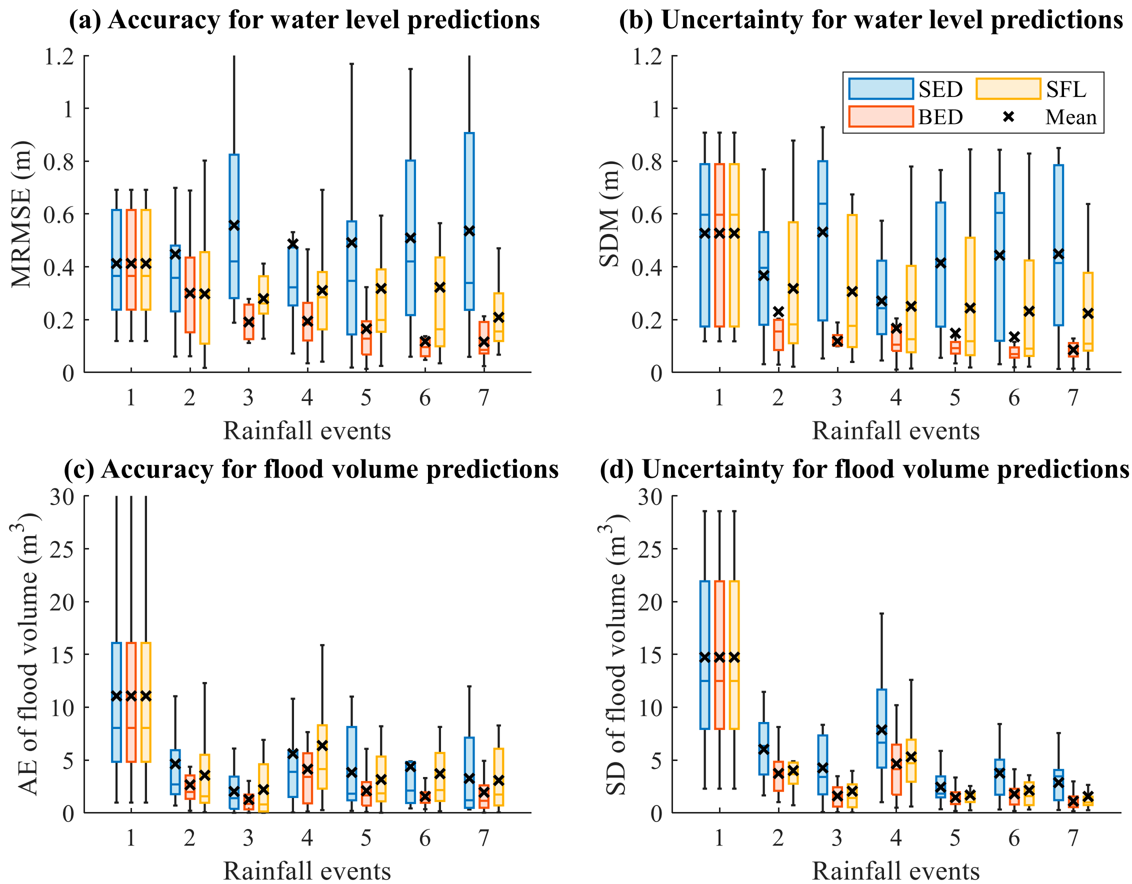

Figure 10. Comparison of assessment metric values for water level predictions (a,b) and flood volume predictions (c,d) at all flood nodes for three design methods.

the SED method. This indicates that calibrated models using the BED method offer the best prediction performance for both flood dynamics and quantities, followed by the SFL method, and then the SED method.

Therefore, based on the overall analysis from Figures 7–10, it can be concluded that the proposed BED method provides more effective observation data for UDM calibration compared to the SED and SFL methods. This significantly enhances the overall prediction performance of calibrated models, particularly in flood prediction, which is of greater practical concern. Please note that, due to space limitations, only the water level prediction results for flood nodes will be presented as representative in the subsequent analysis for flood prediction performance.

4.1.4. Robustness of the proposed method

After the effectiveness of the proposed BED method was verified under parameter scenario P1, its robustness was further tested by applying the method to various parameter scenarios (P2–P5), accounting for the parameter uncertainty in real-world systems. For performance comparison, the SED and SFL methods were also implemented and the average performance of calibrated models across the five scenarios P1–P5 for the three methods was evaluated, as shown in Figure 11. Specifically, Figure 11(a,b) and Figure 11(c,d) present the accuracy and uncertainty results for water level predictions at all nodes and only flood nodes, respectively, while Figure 11(e,f) shows the estimation accuracy and uncertainty of calibration parameters, respectively.

As shown in Figure 11(a–d), for either all nodes or only flood nodes, the BED method achieves the smallest MRMSE and SDM values across rainfall events 2–7, followed by SFL and then SED. This result indicates that the proposed BED method, combined with sequential Bayesian updating, yields calibrated models with the best average prediction performance across different rainfall events and parameter scenarios, demonstrating its robustness for UDM calibration.

Regarding parameter estimation in Figure 11(e–f), the BED method consistently achieves smaller RBM and RBSD values compared to the SFL and SED methods across rainfall events 1–7, indicating the most accurate and least uncertain parameter estimates. Additionally, Figure 11(e–f) shows that parameter uncertainties

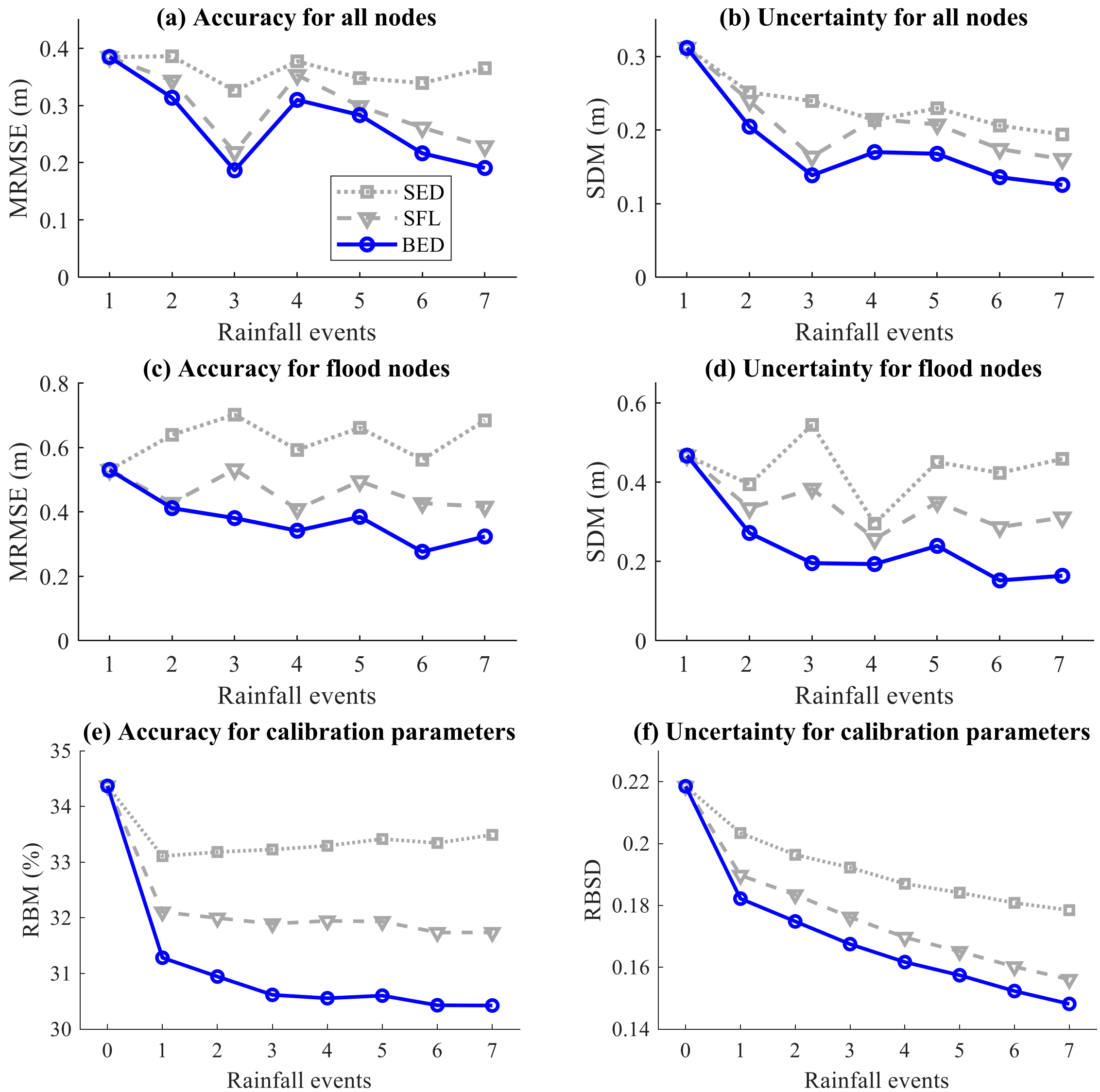

Figure 11. Comparison of the average performance of calibrated models using the three design methods under parameter scenarios P1–P5, focusing on (a,b) water level predictions at all nodes, (c,d) water level predictions at flood nodes, and (e,f) the estimation of calibration parameters.

diminish significantly with sequential updating, while the accuracy of parameter estimates improves only slightly. This further suggests the challenge of identifying true parameter values for UDM calibration, as discussed previously.

4.2. Influence of the spatial density of sensors

Next, the influence of the spatial density of sensors on the effectiveness of model calibration is examined. Five sensor scenarios are considered, involving the placement of 1, 3, 5, 7, and 9 sensors. Figure 12 shows the design results obtained via the BED method; the optimal design for 5 sensors was presented in Figure 4(b), and so is not included here. The figure shows that the BED method can produce optimized sensor locations that achieve both distribution uniformity and maximum data worth for different number of sensors. The SED and SFL methods were also applied to the five sensor scenarios. SED sensor placements conveniently correspond to installing sensors at the terminal node of each partition in Figure 10, while those by the SFL method are simply locations with maximum expected flood volumes evaluated using Monte Carlo simulations for rainfall events H1–H5 in Table 2 based on prior parameter distributions.

Figure 13 presents the average metric values for water level predictions at all nodes, and only at flood nodes, across rainfall events 2–7 at different number of sensors under parameter scenario P1. Prediction results for rainfall events 1 are excluded because prediction at that phase was based on prior parameter distributions (i.e. the model had not been calibrated). In general, for the

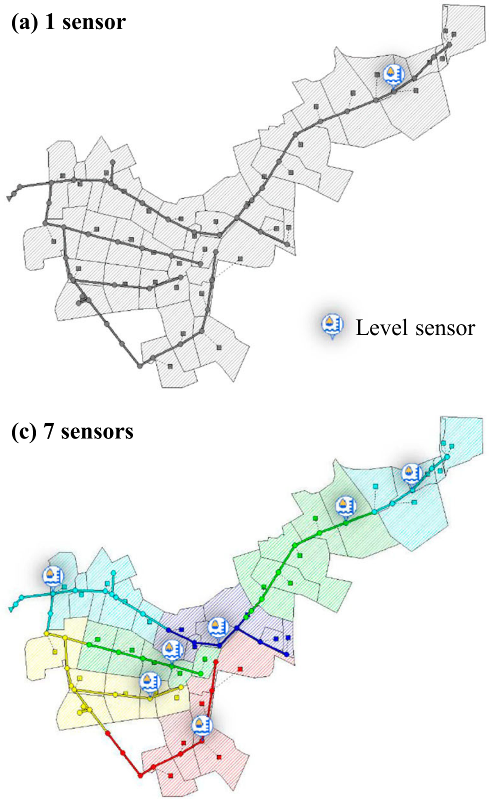

Figure 12. Sensor locations designed by the BED method for different numbers of sensors.

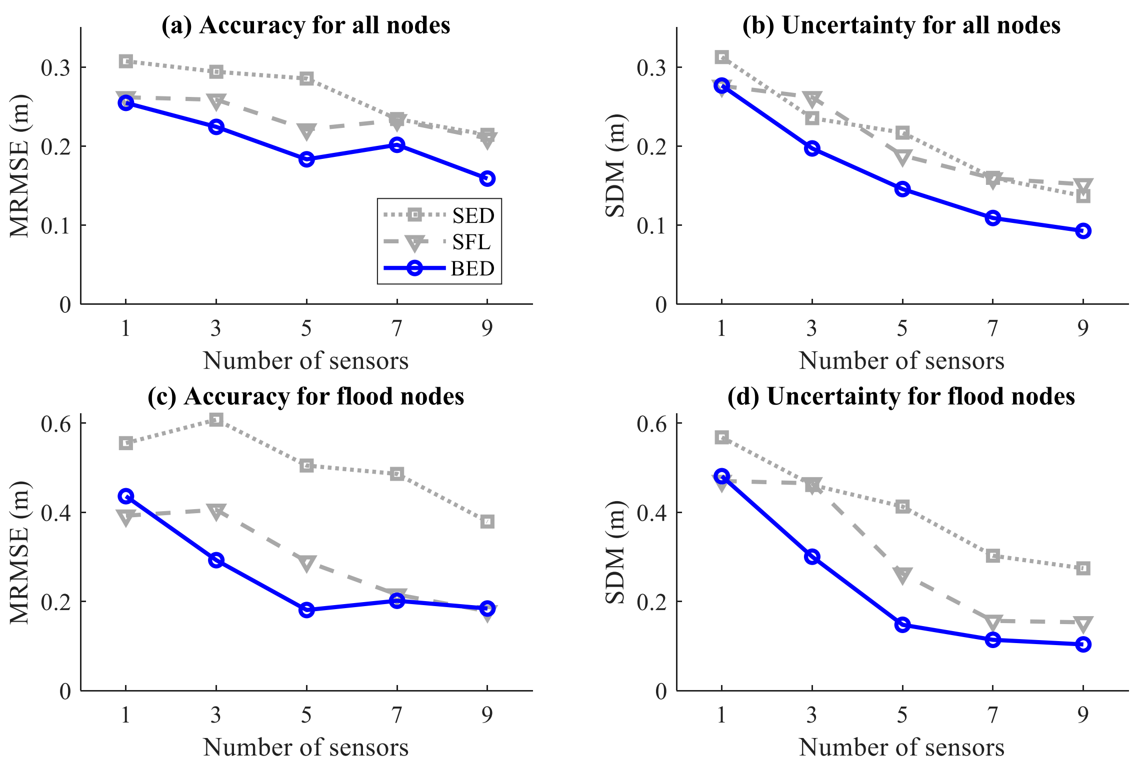

Figure 13. Comparison of the average metric values for water level predictions at (a,b) all nodes, and (c,d) all flood nodes across rainfall events 2–7 for three design methods at different numbers of sensors under parameter scenario P1.

water level predictions at all nodes (Figure 13(a,b)) and at flood nodes (Figure 13(c,d)), the overall predictive performance of calibrated models, and in particular the prediction uncertainty as represented by SDM, improves with the increase in the number of sensors. Clearly, improving the spatial density of sensors can enhance the effectiveness of model calibration.

Overall, BED achieves smaller values of the MRMSE and SDM metrics compared to the SFL method, which in turn achieves smaller values than the SED method. This indicates that BED-based model calibration is the most effective and robust, followed by SFL and then SED. Note, however, that the prediction performance of BED for flood nodes is similar to that of the SFL method for the cases of seven and nine sensors (see Figure 13(c,d)), suggesting that when the number of sensors is adequate (e.g. more than five in this case), the data worth provided by both methods are comparable. The practical implication is that the simpler SFL method (where sensors are placed at locations with a high likelihood of flood) is a viable option for sensor placement when a sufficient number of sensors is available.

4.3. Case 2: placing 10 sensors in a larger UDS

The real-world larger-scale Case 2, which features more complex structures, is next examined. For this case, a total of 1375 unknown parameters (see Table 1) were calibrated. Ten water level sensors were placed within the drainage network for model calibration. The rainfall events in Table 2 were used for Bayesian experimental design and sequential Bayesian updating, as in Case 1.

All three methods (SED, SFL and BED) were implemented to obtain the sensor placement scenarios (see Figure 14). The different colours in Figure 14(a,b) represent the partitioned subregions in Phase 1 of the BED method. Figure 14 shows that three design methods result in significant differences in sensor locations. Sensor locations designed by the SED method (Figure 14(a)) are uniformly distributed while those by the BED method (Figure 14(b)) are further optimized for maximizing the data utility. Sensors designed by the SFL method (Figure 14(c)) are relatively concentrated in the lower part of the network.

Figure 15 compares the average metric values for water level predictions at all nodes and at flood nodes, respectively, for the three design methods under parameter scenarios P1–P5 (see Table 2). For the water level predictions at all nodes (Figure 15(a,b)) and at flood nodes (Figure 15(c,d)), the metric rankings for the three methods are generally BED $<$ SFL $<$ SED, indicating that the BED method has better performance than SFL, which is superior to SED. The only exception is the prediction

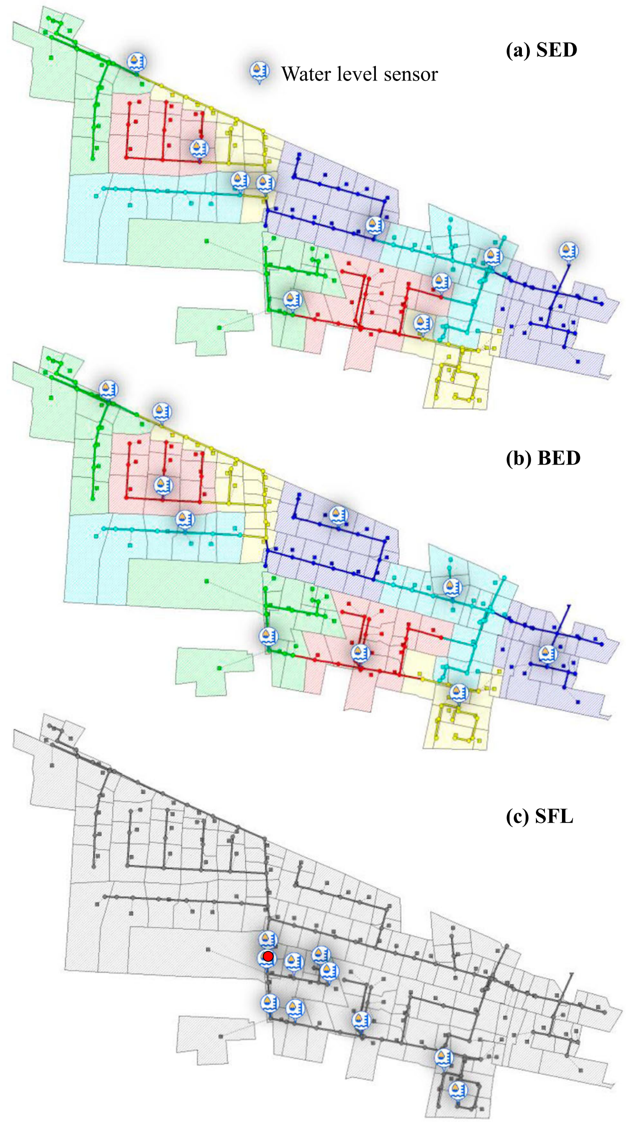

Figure 14. Sensor locations designed by the SED, BED and SFL methods, respectively.

performance for rainfall event 3 as shown in Figure 13(c,d), in which the SFL method is superior to BED. This is because the sensor locations designed by SFL cover almost all flood nodes in this relatively small rainfall event. In general, however, the results support our conclusion that BED-based model calibration combined with sensor placement can achieve superior performance (in terms of both effectiveness and robustness across various rainfall events and parameter scenarios) compared to SFL and SED.

5. Conclusions

This paper addresses the challenge of building an urban drainage model (UDM) with accurate and reliable predictive performance, under realistic conditions where

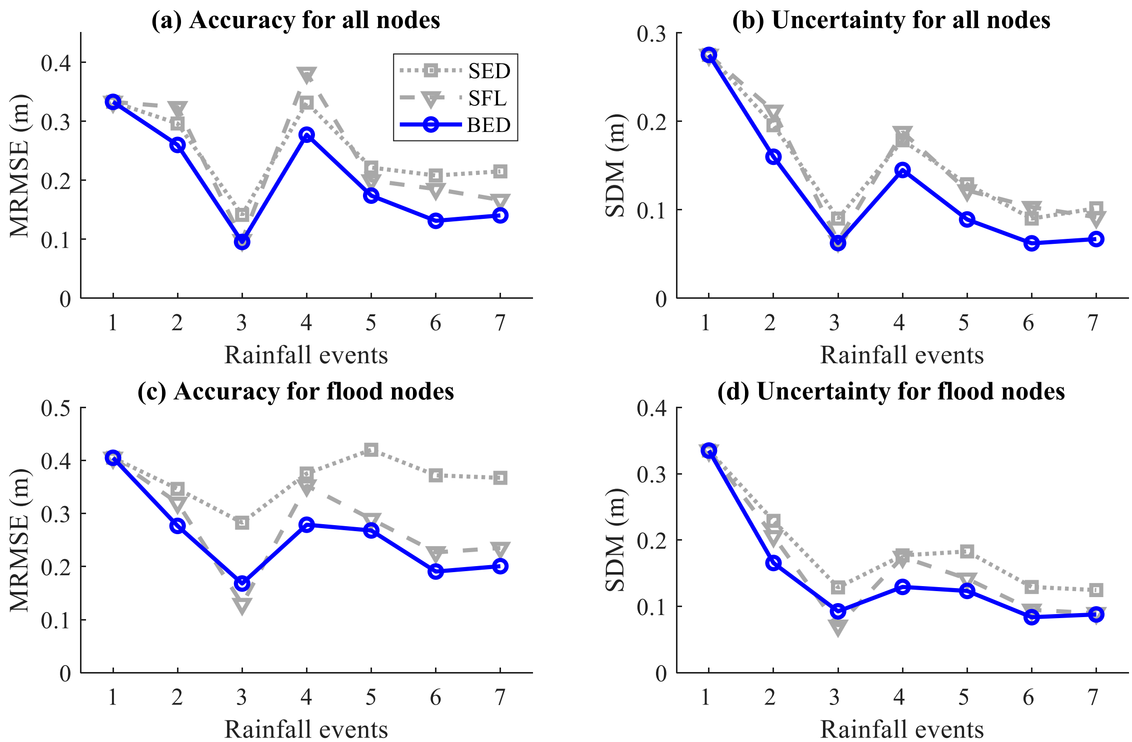

Figure 15. Comparison of the average metric values for water level predictions at (a,b) all nodes, and (c,d) flood nodes for three design methods under parameter scenarios P1–P5.

sensor data is often limited. A novel approach is proposed, based on Bayesian decision theory, to simultaneously optimize sensor placement and calibrate the model for urban drainage pipe systems. Referred to as BED, Bayesian experimental design is combined with a graph partition strategy to optimize sensor placement for model calibration. The graph partition uses a multilevel k-way partitioning scheme to distribute the sensors evenly across partitions, ensuring uniform sensor distribution. The BED method then identifies optimal sensor locations within each partition by maximizing the expected data worth, calculated as the relative entropy from prior to posterior probabilities. Once sensors are deployed, an ensemble smoother with multiple data assimilation (ES-MDA) Bayesian inferential approach is used to sequentially calibrate the model using observation data from sensors responding to chronological rainfall events.

Systematic testing and analysis on two real-world UDMs of different scales reveal that models calibrated with sensors designed by the BED method outperform those using evenly distributed sensors (SED method) or sensors placed in high flood likelihood areas (SFL method), in terms of both prediction accuracy and resulting uncertainty. This performance superiority is evident across the entire model domain and in the flood scenarios that are of great concern in practice, under various rainfall and parameter scenarios. Specifically, the sensor placement results indicate that BED can select optimized sensor locations that consider both distribution uniformity and data worth maximization, while SED produces evenly distributed sensors, and SFL concentrates sensors in areas with severe expected flood volumes.

In conclusion, this paper proposes an effective and robust method that enhances UDM performance for state prediction by simultaneously optimizing sensor placement and model calibration using Bayesian experimental design and Bayesian inference. The approach can be readily applied to actual drainage systems without the need for unrealistic assumptions, and is highly adaptable to various scenarios, such as estimating different parameter types with distinct prior information, utilizing alternative utility functions (instead of relative entropy) to quantify data worth, and selecting observation data of diverse types (e.g. the conduit flow).

The key advantage of the proposed method lies in its ability to integrate sensor placement with model calibration, accounting for uncertainties in all model parameters. This results in a more comprehensive calibration framework compared to traditional methods, leading to more accurate and stable model predictions, particularly in complex urban drainage systems. However, the current method assumes a fixed number of sensors, which limits its flexibility. Future research can expand this into a multi-objective optimization framework, enabling the optimal number of sensors to be determined by balancing model accuracy and sensor deployment costs.

The proposed method has diverse applications in urban flood management, including real-time flood prediction, early warning systems, and the optimization of stormwater management infrastructure. Looking ahead, an important area for improvement would be the integration of water quality monitoring into the framework. Addressing contaminants, such as conservative or degradable substances, would extend the method's applicability to urban sewage systems, where monitoring pollutants and detecting critical events like industrial spills are essential (e.g. Sambito et al., 2020, 2022; Sambito & Freni, 2021). Given the growing importance of urban sustainability and environmental monitoring, combining flood and water quality sensor placement would provide a more comprehensive approach to urban drainage management, addressing both flood risks and long-term environmental challenges.

Acknowledgements

The authors would acknowledge Agnethe Nedergaard Pedersen from DTU Environment, Denmark for providing the open data and models for community-wide urban drainage systems research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was supported by the National Key R&D Program of China [grant numbers 2023YFC3208901, 2023YFC3208900], the National Natural Science Foundation (NSFC) of China [grant numbers 52000156 and 52179080], the NSFC/RGC Joint Research Scheme [grant number 52261160379], and the Fundamental Research Funds for the Central Universities [grant number B210201048].

Data availability statement

The data that support the findings of this study are openly available in Figshare at https://figshare.com/s/b4865490d310efe00e55.

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